

University of Al-Kafeel

College of Dentistry



LEC-3

Cholinergic Agonist Drugs

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Learning Objectives

Pharmacology For Dentistry

By the end of this lecture, you should be able to



- Describe the Mechanism of Action.
- Differentiate Between Direct and Indirect Cholinergic Agonists.
- Identify Therapeutic Uses and Clinical Indications.
- Recognize Adverse Effects and Contraindications

Cholinergic agonists

- Cholinergic agonists are the name given to a group of medicines that **mimic the actions of acetylcholine**.
- Acetylcholine is one of the most common neurotransmitters in our body, and it has actions in both the central and peripheral nervous systems.
- Drugs produce effects similar to that of ACh called (**Cholinomimetic Drugs**) or (**Cholinergic Agonist**)
- Cholinergic agonists activate receptors that respond to acetylcholine, especially in the parasympathetic nervous system, which controls "rest and digest" functions like: Salivation Digestion, Urination. Lacrimation (tear production)
- Drugs produce effects opposite to that of ACh called (**Anticholinergic Drugs**) or (**Cholinergic Antagonist**)



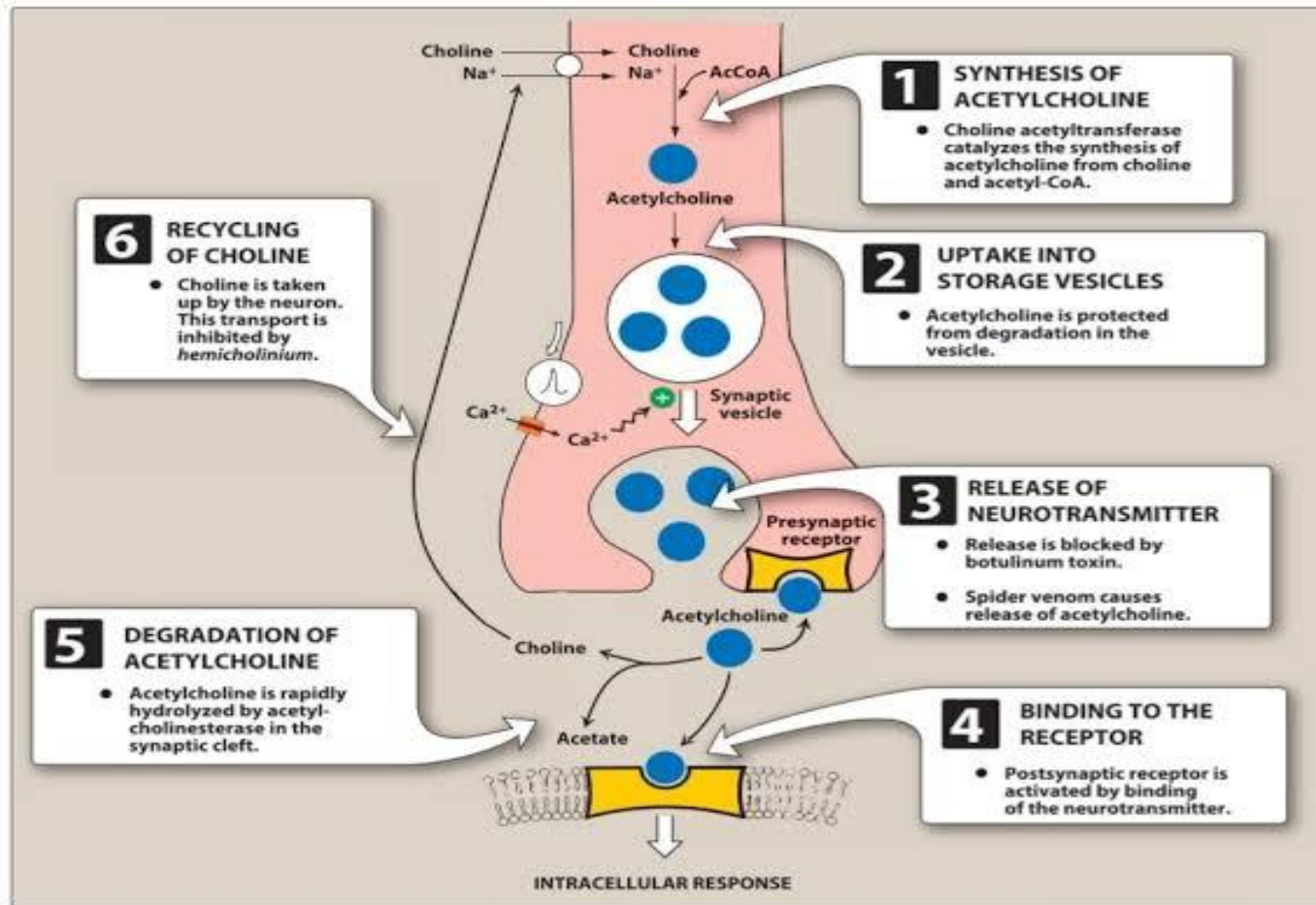
The Six Steps of Cholinergic Neurotransmission



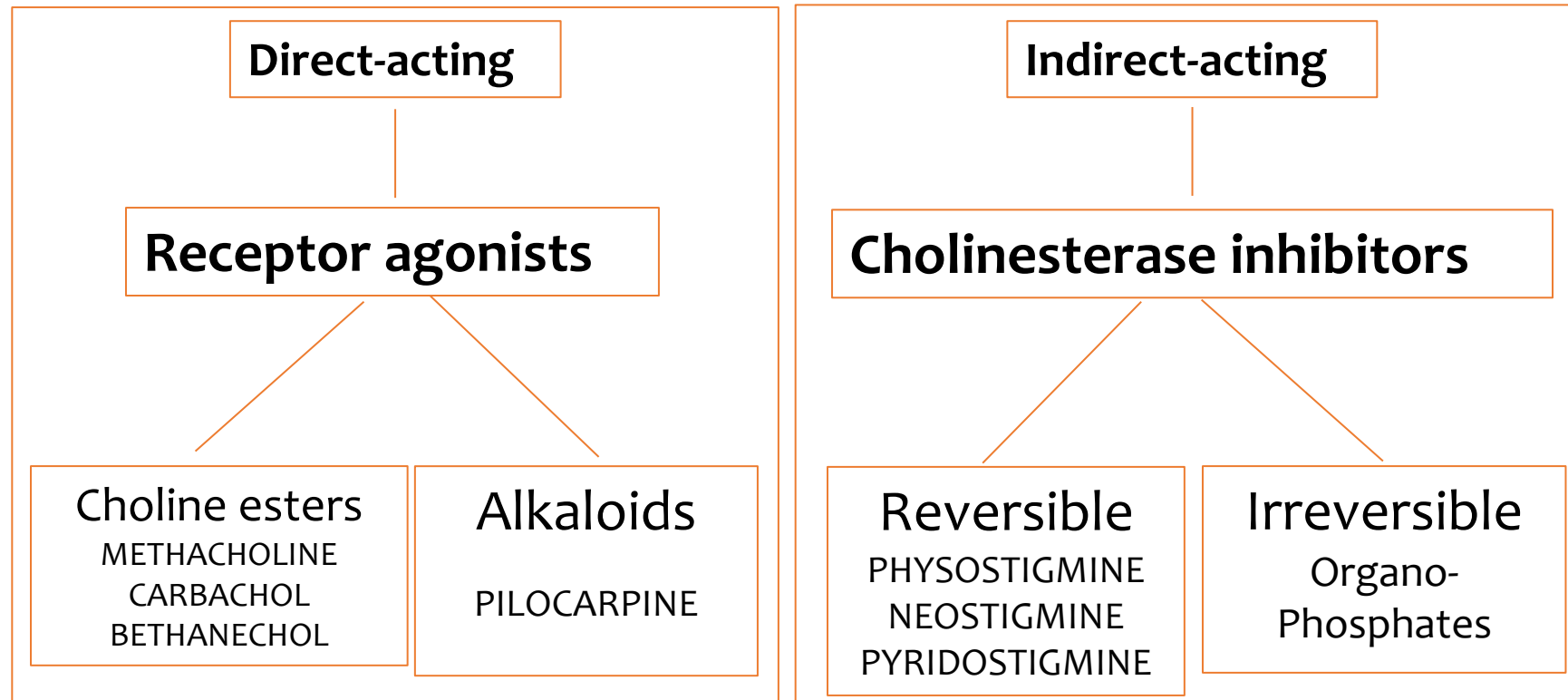
Step	Main Process	Key Points (Simplified Mechanism)	Drugs / Toxins That Block This Step
1. Synthesis of Acetylcholine (ACh)	Production of ACh from choline + acetyl-CoA	Choline is transported into the neuron (Na^+ -dependent), then combined with acetyl-CoA to form ACh.	Hemicholinium-3 – blocks choline uptake into the neuron (rate-limiting step).
2. Storage of Acetylcholine	ACh packaged into vesicles	ACh is loaded into vesicles for protection and release control.	Vesamicol – blocks ACh transport into synaptic vesicles.
3. Release of Acetylcholine	ACh released into synaptic cleft (exocytosis)	Action potential $\rightarrow \text{Ca}^{2+}$ influx $\rightarrow$ vesicle fusion $\rightarrow$ ACh release.	Botulinum toxin – prevents ACh vesicle fusion/release. Black widow spider toxin (α-latrotoxin) – causes excessive ACh release.
4. Binding to Postsynaptic Receptors	ACh binds to nicotinic or muscarinic receptors	ACh activates receptors: Nicotinic (fast ion channel) and Muscarinic (slow, G-protein linked).	Atropine – blocks muscarinic receptors. Curare (tubocurarine) – blocks nicotinic receptors at neuromuscular junction.
5. Degradation of Acetylcholine	Breakdown of ACh into choline + acetate	ACh is rapidly destroyed to stop stimulation and allow reset.	Organophosphates (e.g., sarin, malathion) and neostigmine – inhibit ACh breakdown (increase ACh levels).
6. Recycling of Choline	Choline reuptake into presynaptic neuron	Choline from ACh breakdown is reabsorbed for reuse in synthesis.	Hemicholinium-3 – also blocks choline reuptake, slowing ACh resynthesis.



The Six Steps of Cholinergic Neurotransmission



Cholinomimetic drugs



Pharmacological actions of Cholinergics

Effector organs	Effects
Eye	Myosis Accommodation Reduction of intraocular pressure (IOP) Increase lacrymation.
Heart	↓ heart rate ↓ force of contraction Slow conduction
Blood vessels	Constriction of coronary arteries
Stomach and Intestine	↑ motility and secretion
Urinary bladder	Relaxation of sphincters and contraction of bladder wall
Respiratory system	Constriction of bronchi increased bronchial secretions
Exocrine glands	Stimulate all e.g. sweating, salivation and lacrymation.



A-CHOLINE ESTERS

Choline Ester	Susceptibility to Cholinesterase	Muscarinic Action	Nicotinic Action	Clinical Use	Major Side Effects
Acetylcholine	++++	+++	+++	Rarely used clinically (short duration); sometimes used in eye surgery to produce miosis (e.g., intraocular use during cataract surgery)	Bradycardia, hypotension, bronchospasm, diarrhea, sweating, salivation
Methacholine	+	++++	None	Diagnosis of bronchial hyperreactivity in asthma testing (Methacholine challenge test)	Bronchospasm, hypotension, bradycardia—testing should be done under medical supervision
Carbachol	Negligible	++	+++	Glaucoma (reduces intraocular pressure by miosis) miosis during eye surgery	Blurred vision, headache, sweating, GI cramps, hypotension (if absorbed systemically)
Bethanechol	Negligible	++	None	Postoperative ileus and urinary retention (stimulates GI and bladder motility)	Sweating, salivation, flushing, hypotension, diarrhea, abdominal cramps, bronchospasm

A-CHOLINE ESTERS

• **Bethanechol:**

- selective (muscarinic) and not hydrolyzed by cholinesterases. It acts mainly on M3 receptors and has little effect on the heart.

➤ **Clinical use of Bethanechol:**

Bladder and bowel atony, for example, after surgery or spinal cord injury,
congenital megacolon
post-operative urine retention, postpartum urinary retention.

➤ **adverse effects:**

All parasympathomimetic effects: cyclospasm, diarrhea, urinary urgency, plus vasodilation, reflex tachycardia, and sweating, salivation.

A-CHOLINE ESTERS

Carbachol

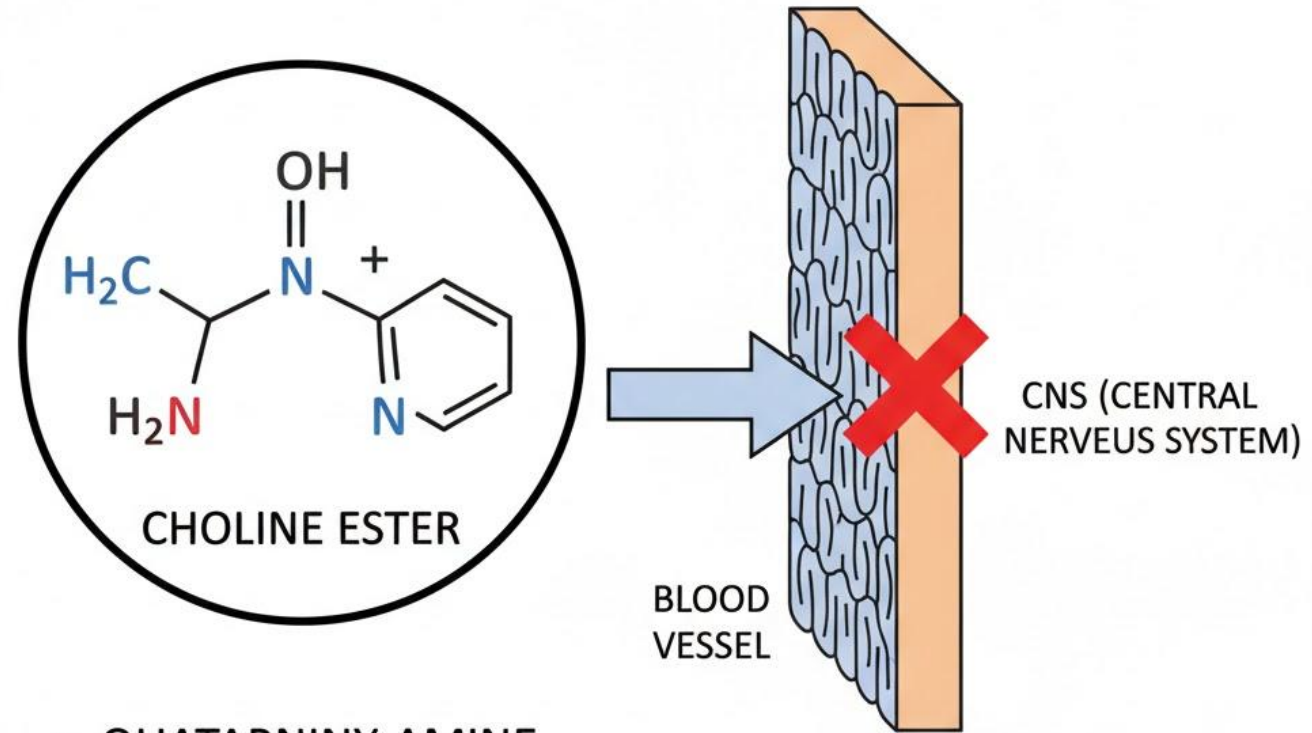
Activates muscarinic (M) and nicotinic (N) receptors.
Used locally in treatment of glaucoma (*pilocarpine is preferable*)

Methacholine

Used to test bronchial hyper-reactivity (Diagnosis of asthma)

A-CHOLINE ESTERS

All choline esters are quaternary amines, have +ve charge, and can't cross the BBB to enter the CNS



- QUATARNINY AMINE
- +VE CHARGE
- CAN'T CROSS BBB INTO CNS

B-ALKALOIDS

Pilocarpine:

Pharmacological effects:

On the eye:

- Rapid **miosis** (contraction of the **circular muscle of the iris**)
- Contraction of **ciliary muscles** (**accommodation for near vision**).

This will result in opening the **trabecular meshwork** around **Schlemm's canal**, causing an immediate drop in intraocular pressure (**IOP**) as a result of the increased drainage of aqueous humor.

Other effects: Pilocarpine is one of the most potent **stimulators of secretions** such as sweat, tears, and saliva



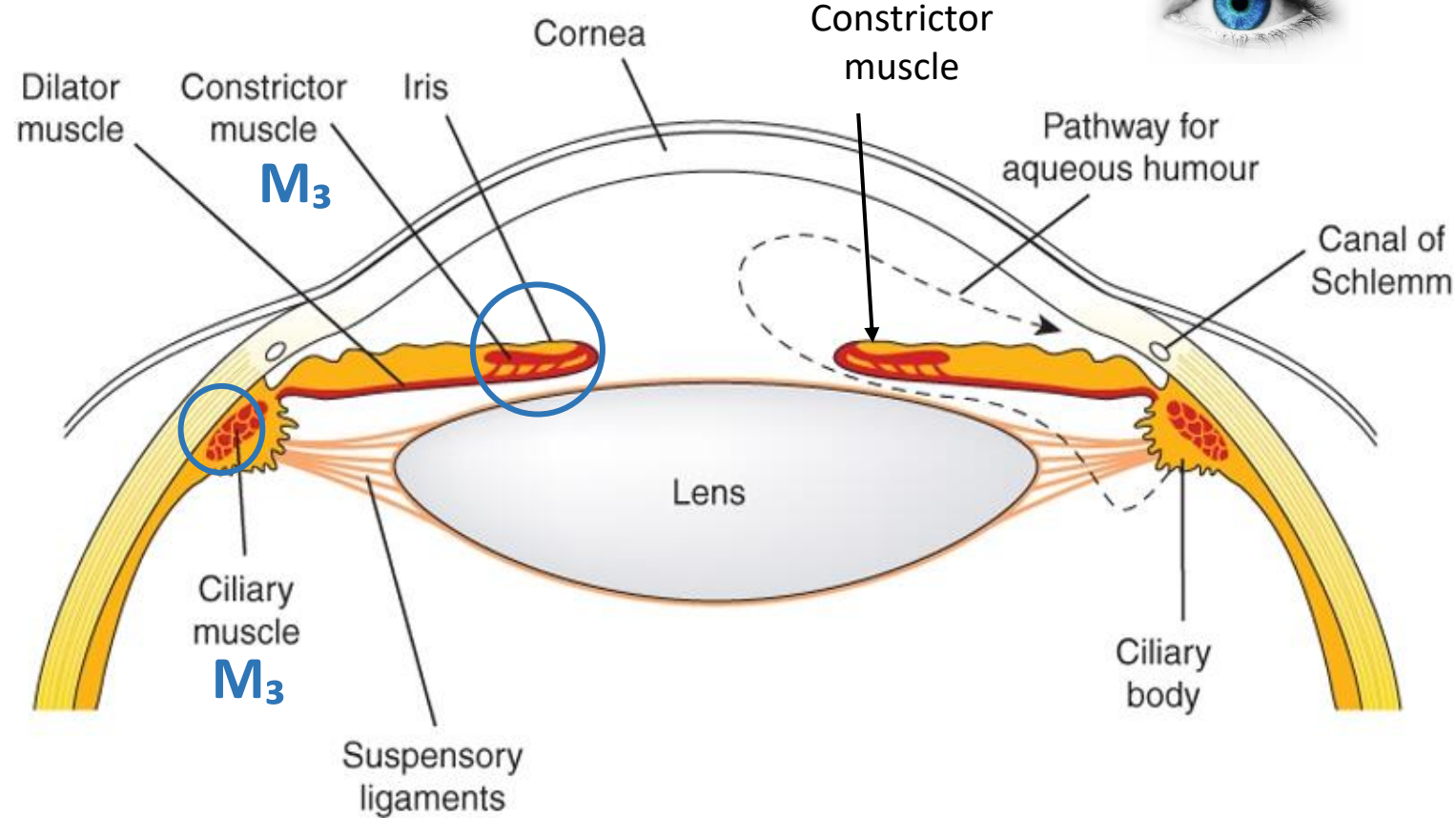
Pilocarpine:

Therapeutic uses:

- ***Glaucoma***: pilocarpine is the drug of choice in the emergency lowering of ***IOP***. It is **used topically** (eye drops) for this purpose.
- ***Xerostomia*** (dryness of the mouth): pilocarpine is **used orally** in patients with xerostomia resulting from irradiation of the head and neck tumours.
- ***Sjögren's syndrome*** (increases salivation).

SE: CNS disturbances, profuse sweating, salivation & lacrimation

Effects of Pilocarpine the eye



Parasympathetic stimulation: Contraction of circular muscle produces constriction (miosis)
Contraction of ciliary muscle produces accommodation for near vision

Cholinesterase inhibitors (Anticholinesterases)

Inhibit the action of **AChE enzyme**, resulting in **accumulation of ACh** at muscarinic and nicotinic sites and this will lead to **intensifying ACh effects**. According to their **binding** to the enzyme the inhibition could be **reversible** or **irreversible**.

Cholinesterase Enzymes (two distinct types):

- 1. Acetylcholinesterase** (found in *neural tissue & RBC*)
 - 2. Butyrylcholinesterase** or **pseudocholinesterase** (found in **plasma, liver, skin, brain and GIT**)
- Anticholinesterases **inhibit both forms** of the enzyme

THE INDIRECT-ACTING CHOLINESTERASE INHIBITORS

have many effects like those of the direct-acting agonists, but they act by inhibiting the enzyme that terminates the action of endogenous acetylcholine.

❑ **Reversible anticholinesterases**

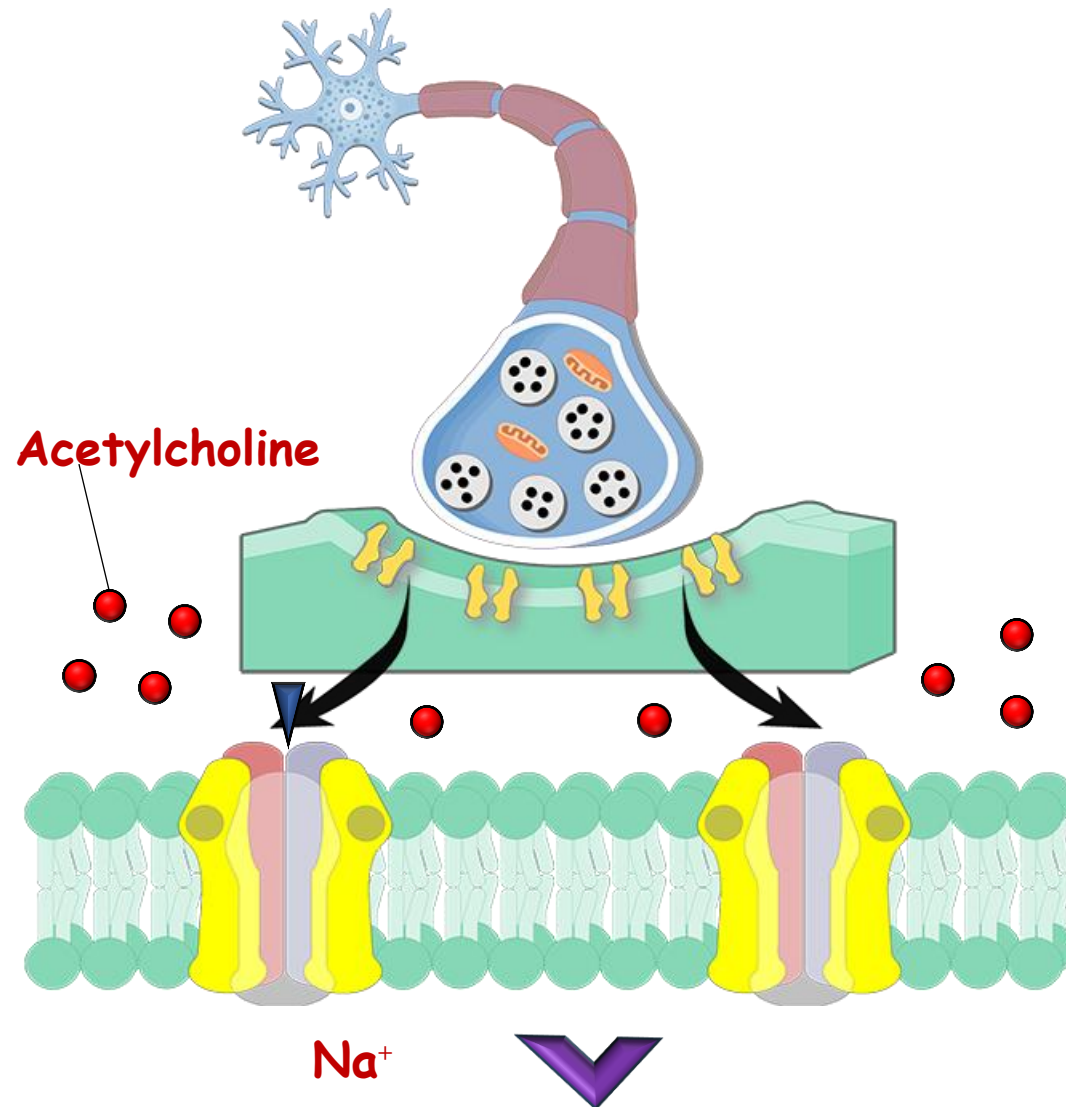
Very short-acting anticholinesterases

Edrophonium

- The action of edrophonium is similar to physostigmine, except that it is more rapidly absorbed and has short duration of action (10-20 min).
- Used in diagnosis of myasthenia gravis (I.V injection lead to rapid increase in muscle strength)

Drug / Class	Action	Indications	Pharmacokinetics	Side Effect
Neostigmine	Reversible inhibitor of cholinesterase	- Reversal of neuromuscular junction (NMJ) block caused by tubocurarine - Treatment of myasthenia gravis	- Quaternary compound (more polar) - Does not cross the blood–brain barrier (no CNS effect) - Duration: 2–4 hours	Muscarinic side effects (bradycardia, salivation, diarrhea, bronchospasm)
Physostigmine	Reversible inhibitor of cholinesterase	- Antidote for anticholinergic (antimuscarinic) poisoning (e.g., atropine, TCA toxicity) - Treatment of atony of bladder and paralytic ileus - Topical use in glaucoma (induces miosis)	- Tertiary amine (lipid soluble) — crosses BBB - Duration: 2–8 hours	CNS effects (convulsions, restlessness, bradycardia, hypotension)
Pyridostigmine	Reversible inhibitor of cholinesterase	- Chronic management of myasthenia gravis - Reversal of nondepolarizing NMJ blockade	- Quaternary compound (limited CNS penetration) - Duration: 3–6 hours	Similar side effects as neostigmine; longer acting and better tolerated
Organophosphorus Compounds (e.g., Parathion, Malathion, Sarin, Echothiophate)	Irreversible inhibitors of cholinesterase — phosphorylate enzyme active site → long-lasting inactivation	- Echothiophate (topical): chronic glaucoma treatment (rarely used) - Other OP compounds: insecticides (malathion, parathion) or nerve agents (sarin, soman)	- Highly lipid soluble (except echothiophate) - Very long duration (enzyme regeneration takes days–weeks)	- Severe cholinergic crisis (miosis, bronchorrhea, convulsions, paralysis, death) Antidote: Atropine (blocks muscarinic receptors) + Pralidoxime (2-PAM) (reactivates cholinesterase if given early)

myasthenia gravis



THE INDIRECT-ACTING CHOLINESTERASE INHIBITORS

Reversible anticholinesterases

Intermediate -acting anticholinesterases

carbonic acid ester

Neostigmine

Action:

- Reversible Inhibitor of cholinesterase.

Indications:

- Antidote to reversal of NMJ block caused by **tubocurarine**
- Treatment of myasthenia gravis.

Pharmacokinetics:

- More polar and Do not penetrate the CNS
- 2-4 hours



Physostigmine

Action:

- Reversible Inhibitor of cholinesterase.

Indications:

- Antidote for drugs with anticholinergic action such as **atropine** (antimuscarinic)
- Treatment Atony of bladder and paralytic ileus.
- occasionally used in acute glaucoma (topical) (induce miosis and spasm of accommodation).

Pharmacokinetics:

- Lipid soluble and have CNS activity.
- can be used topically in the eye, 2-8 hours

Pyridostigmine

Action:

- Reversible Inhibitor of cholinesterase.

Indications:

- Chronic Management of Myasthenia gravis.

Pharmacokinetics:

- Lipid soluble and penetrate the CNS
- 3-6 hours



THE INDIRECT-ACTING CHOLINESTERASE INHIBITORS

intermediate-acting anticholinesterases

• Physostigmine

Natural alkaloid tertiary amine derived from *Physostigma Veneosum* plant. It inhibits cholinesterase enzymes reversibly. This will **delay the hydrolysis of ACh until the AChE became free**. The result is the **potentiation of cholinergic activity throughout the body**.

- **Muscarinic** and **nicotinic** effects (wide effects in the body)
- **CNS effects** because they cross the BBB.

THE INDIRECT-ACTING CHOLINESTERASE INHIBITORS

- **Physostigmine**

- **Action:** Inhibitor of cholinesterase; amplifier of endogenously released ACh
- **Indications:**
 - ❖ Reversal of severe atropine poisoning (IV);
 - ❖ occasionally used in acute glaucoma (topical) (induce miosis and spasm of accommodation)
 - ❖ Used in intestinal atony (increases their motility)
- **Pharmacokinetics:** Lipid soluble; can be used topically in the eye. Duration of action: 2–4 h
- **Adverse effect:** Generalized cholinergic stimulation plus CNS effects: seizures



THE INDIRECT-ACTING CHOLINESTERASE INHIBITORS

Neostigmine

A **synthetic** compound that is a **quaternary amine**, hence, it is more polar (water soluble) and **not enter the CNS**.

Indications:

- Treatment of **Myasthenia Gravis** (nicotinic action).
Myasthenia Gravis is an **autoimmune disease** in which there are antibodies against **NMJ** resulting in a **decrease in the No. of nAChRs** and in the **No. of ACh vesicles** in the nerve terminal. The disease is characterized by **muscle weakness** starting in the face & neck and the progressive stage involves limbs and intercostal muscles.
- Termination of action of **Curare** and other non-depolarizing blockers of **NMJ** (muscle relaxants) during **general anaesthesia** (nicotinic action).
- Also used for its **muscarinic action** in (**atony of bladder & GIT**)

When the nicotinic effect of the drug is wanted (as in the 1st & 2nd indications) **muscarinic side effects can be blocked by atropine** (antimuscarinic drug)

THE INDIRECT-ACTING CHOLINESTERASE INHIBITORS

Pyridostigmine:

Used in the **chronic management of myasthenia gravis**. Preferred to neostigmine because it has a longer duration of action.



Organophosphorus

Organophosphorous compounds bind **covalently** to **AChE** to produce the **phosphorylated enzymes**.

The result is a **long-lasting accumulation** of ACh at all sites. Many of these agents are **extremely toxic** and were developed by the military as **nerve agents**. { **Tabun** (GA), **Sarin** (GB) and **Soman** (GD) are gases, while **VX** is a thick liquid }

Compounds, such as **parathion** & **malathion** are employed as **organophosphorus insecticides** used to control insects in agricultural practice.

Very few compounds are used as **drugs** like **Ecothiopate** which is used topically in the treatment of glaucoma (**never used systemically**)

Organophosphorus Toxicity

- Toxicity by **organophosphorus** is due to
A-Muscarinic effects (**DUMBLES**) diarrhea, urination, miosis, bronchospasm, bradycardia, lacrimation, emesis, salivation, and sweating.
B-nicotinic effects (muscle twitches, neuromuscular blockade of diaphragm and intercostal muscles).
C-CNS effects (restlessness, insomnia, tremors, confusion, convulsions, coma, death from respiratory failure).

Management of Organophosphorus Toxicity



- the **antidote** of **first choice** is the antimuscarinic agent **atropine**, but this drug has **no effect** on the **nicotinic signs of toxicity**.
- **Nicotinic toxicity** is treated by **regenerating active cholinesterase**. Immediately after binding to cholinesterase, most organophosphate inhibitors can be removed from the enzyme by the use of regenerator compounds such as **pralidoxime** and this may reverse both **nicotinic and muscarinic signs**
- If the enzyme-phosphate binding is allowed to persist, however, aging (a further chemical change) occurs, and regenerator drugs can no longer remove the inhibitor.



References



- **Lippincott's** Illustrated Reviews: Pharmacology, 9th Edition. 2023
- **Katzung's** Basic & Clinical Pharmacology, 14th Edition. 2018
- **Rang and Dale's** Pharmacology, 8th Edition. 2021



THANK
YOU!

